

6 Proof by Contradiction

Use the method of proof by contradiction to prove the following statements. (In each case, you should also think about how a direct or contrapositive proof would work. You will find in most cases that proof by contradiction is easier.)

2. Suppose $n \in \mathbb{Z}$. If n^2 is odd, then n is odd.

Proof. For the sake of contradiction, suppose n^2 is odd and n is not odd. Then n^2 is odd and n is even. Since n is even, $n = 2c$ for some $c \in \mathbb{Z}$. Then $n^2 = (2c)^2 = 4c^2 = 2(2c^2)$, so n^2 is even. Thus n^2 is odd and n^2 is even, a contradiction. $\square$

4. Prove that $\sqrt{6}$ is irrational.

Proof. For the sake of contradiction, suppose $\sqrt{6}$ is not irrational. Then $\sqrt{6}$ is rational. Then $\sqrt{6} = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$, where $\frac{a}{b}$ is fully reduced; in particular, a and b are not both even. Squaring both sides gives $6 = \frac{a^2}{b^2}$, thus $a^2 = 6b^2 = 2(3b^2)$. Then a^2 is even, which implies a is even. Since a and b are not both even, it follows that b is odd, and hence b^2 is odd. Since a is even, $a = 2k$ for some $k \in \mathbb{Z}$. Substituting this value for a in the equation we get $(2k)^2 = 2(3b^2)$, so $4k^2 = 2(3b^2)$, and dividing both sides by 2 gives $2k^2 = 3b^2$. Thus $3b^2$ is even. But b^2 is odd and 3 is odd, and the product of two odd numbers is odd, so $3b^2$ is odd. Thus $3b^2$ is both even and odd, a contradiction. $\square$

6. If $a, b \in \mathbb{Z}$, then $a^2 - 4b - 2 \neq 0$.

Proof. For the sake of contradiction, suppose $a, b \in \mathbb{Z}$ but $a^2 - 4b - 2 = 0$. Then we have $a^2 = 4b + 2 = 2(2b + 1)$, which means a^2 is even. Therefore a is even, so $a = 2c$ for some $c \in \mathbb{Z}$. Substituting this value for a back in the equation we get

$$(2c)^2 - 4b - 2 = 0$$

$$4c^2 - 4b - 2 = 0$$

$$4(c^2 - b) = 2$$

$$2(c^2 - b) = 1$$

Since $c^2 - b \in \mathbb{Z}$, the last equation says that 1 is an even number, a contradiction. $\square$

8. Suppose $a, b, c \in \mathbb{Z}$. If $a^2 + b^2 = c^2$, then a or b is even.

Proof. For the sake of contradiction, suppose that $a^2 + b^2 = c^2$, and that a and b are odd. Then there exist integers m and n for which $a = 2m + 1$ and $b = 2n + 1$. Hence, $a^2 = (2m + 1)^2 = 4(m^2 + m) + 1$, so $a^2 \equiv 1 \pmod{4}$. Similarly $b^2 \equiv 1 \pmod{4}$. Since $a^2 + b^2 = c^2$, it follows that $c^2 \equiv 1 + 1 \equiv 2 \pmod{4}$. But every square is congruent to either 0 or 1 $\pmod{4}$, so $c^2 \not\equiv 2 \pmod{4}$, a contradiction. $\square$

10. There exist no integers a and b for which $21a + 30b = 1$.

Proof. For the sake of contradiction, assume there exist integers a and b for which $21a + 30b = 1$. Then $1 = 3(7a + 10b)$. Since $7a + 10b \in \mathbb{Z}$, this implies $3 \mid 1$, a contradiction. $\square$

12. For every positive $x \in \mathbb{Q}$, there is a positive $y \in \mathbb{Q}$ for which $y < x$.

Proof. For the sake of contradiction, assume there exists a positive $x \in \mathbb{Q}$ such that every positive $y \in \mathbb{Q}$ satisfies $y \geq x$. Consider $y = \frac{x}{2}$. Since x is rational, $x = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$, so $\frac{x}{2} = \frac{a}{2b}$ with $2b \neq 0$; hence $\frac{x}{2}$ is rational. Since $x > 0$, we have $\frac{x}{2} > 0$ and $\frac{x}{2} < x$. Thus y is a positive rational number with $y < x$, a contradiction. $\square$

14. If A and B are sets, then $A \cap (B - A) = \emptyset$.

Proof. For the sake of contradiction, assume A and B are sets, but $A \cap (B - A) \neq \emptyset$. Then there exists an element $x \in A \cap (B - A)$. Therefore $x \in A$ and $x \in (B - A)$. But if $x \in (B - A)$ then $x \in B$ and $x \notin A$. Thus we have $x \in A$ and $x \notin A$, a contradiction. $\square$

16. If a and b are positive real numbers, then $a + b \geq 2\sqrt{ab}$.

Proof. For the sake of contradiction, assume $a, b \in \mathbb{R}^+$ but $a + b < 2\sqrt{ab}$. Since $a, b > 0$, both sides of the inequality are positive, so squaring preserves the inequality: $a^2 + 2ab + b^2 < 4ab$. Subtracting $4ab$ from both sides, we get $a^2 - 2ab + b^2 < 0$. By factoring the left side, we arrive at $(a - b)^2 < 0$. But the square of a real number is nonnegative, so $(a - b)^2 \geq 0$, a contradiction. $\square$

18. Suppose $a, b \in \mathbb{Z}$. If $4 \mid (a^2 + b^2)$, then a and b are not both odd.

Proof. For the sake of contradiction, assume $4 \mid (a^2 + b^2)$ and a and b are odd. Then there exist integers m and n for which $a = 2m + 1$ and $b = 2n + 1$. Hence $a^2 = (2m + 1)^2 = 4(m^2 + m) + 1$, so $a^2 \equiv 1 \pmod{4}$. Similarly $b^2 \equiv 1 \pmod{4}$. From the definition of divisibility, we know $a^2 + b^2 = 4k$ for some integer k . So it follows that $4k \equiv 1 + 1 \equiv 2 \pmod{4}$. But $4k = 4k + 0$, so $4k \equiv 0 \pmod{4}$. Thus $4k$ is congruent to both 0 and 2 $\pmod{4}$, which is impossible since $4 \nmid 2$, a contradiction. $\square$